

SAT UZBEKISTAN

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1.

The function h is defined by $h(x) = -\sqrt{x^2 + bx + c}$, where b and c are constants. In the xy -plane, the graph of $y = h(x)$ contains the points $(2, 0)$ and $(0, -\sqrt{334})$. If $h(m) = 0$, what is the greatest possible value of m ?

2.

What value of w is the solution to the equation $0.8w - 0.56 = 8(w - 0.001)$?

Answer Preview:

3.

A gym charges its members a onetime \$37 enrollment fee and a membership fee of \$17 per month. If there are no charges other than the enrollment fee and the membership fee, after how many months will a member have been charged a total of \$224 at the gym?

4.

At a convention center, there are a total of 325 visitors. Each visitor is located in either room A, room B, or room C. If one of these visitors is selected at random, the probability of selecting a visitor who is located in room A is 0.44, and the probability of selecting a visitor who is located in room B is 0.36. How many visitors are located in room C?

(A) 20

(B) 26

(C) 51

(D) 65

5.

The function f defined by $f(t) = 11t + 8$ gives the estimated length, in inches, of a pothos plant t months after Malcolm purchased it. Which of the following is the best interpretation of 8 in this context?

- (A) The estimated length of the pothos plant was 8 inches when Malcolm purchased it.
- (B) The pothos plant is expected to grow to a maximum length of 8 inches.
- (C) The pothos plant is expected to grow 8 inches each month.
- (D) Malcolm will keep the pothos plant for 8 months.

6.

Mass (grams)	Frequency
10	12
20	6
30	8
40	9

The frequency table summarizes the mass, in grams, of 35 objects. What is the mean mass, in grams, of these objects?

Answer Preview:

7. The value of a share of stock was \$410 on the day it was purchased. Every year, the value of the share of stock increases by 4% of its value the previous year. Which function V gives the value, in dollars, of the share of stock t years after the day it was purchased?

(A) $V(t) = 410(1.04t)$

(B) $V(t) = 410(1.4t)$

(C) $V(t) = 410(1.04)^t$

(D) $V(t) = 410(1.4)^t$

8.

Line p in the xy -plane contains the points $(6, 0)$ and $(6, 6)$. Which equation defines line p ?

☐ (A) $x = 0$

☐ (B) $x = 6$

☐ (C) $y = 0$

☐ (D) $y = 6$

9.

If $x = \sqrt[n]{2x^n + 35}$, where n is a positive integer constant, what is the value of x^n ?

Answer Preview:

10.

A computer repair specialist charges \$160 for the first two hours of repair plus an hourly fee for each additional hour. The total cost for 6 hours of repair is \$360. Which function f gives the total cost, in dollars, for x hours of repair, where $x \geq 2$?

(A) $f(x) = 50x + 60$

(B) $f(x) = 50x + 160$

(C) $f(x) = 60x$

(D) $f(x) = 60x + 160$

11. A circle in the xy -plane has center $(36, 4)$ and y -intercepts $(0, -23)$ and $(0, 31)$. What is the radius of this circle?

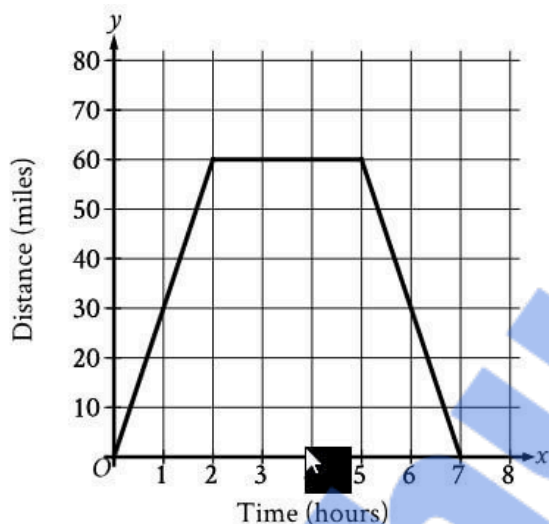
(A) 23

(B) 27

(C) 31

(D) 45

12.



Mohammed drove his car on a straight road to a certain location. The graph models Mohammed's distance, in miles, from his home as a function of time, in hours. Which of the following could the model describe?

- (A) Mohammed drove away from his home at a constant speed for 2 hours, spent 3 hours at the location, then continued to drive away from his home at a constant speed for 2 hours.
- (B) Mohammed drove away from his home at a constant speed for 2 hours, spent 3 hours at the location, then drove back to his home at a constant speed for 2 hours.
- (C) Mohammed drove away from his home at an increasing speed for 2 hours, spent 3 hours at the location, then drove back to his home at a decreasing speed for 2 hours.
- (D) Mohammed drove away from his home at an increasing speed for 2 hours, drove at a constant speed for 3 hours, then drove back to his home at a decreasing speed for 2 hours.

13. A rectangular poster has an area of 300 square inches. A copy of the poster is made in which the length and width of the original poster are each increased by 20%. What is the area of the copy, in square inches?

Answer Preview:

14.

A right rectangular prism has a base area of $56t$ square centimeters (cm^2). The length of the base of the rectangular prism is $\frac{14}{3}$ cm, and the height of the rectangular prism is 3 cm. Which expression represents the surface area, in cm^2 , of the right rectangular prism?

(A) $112t + 56$

(B) $184t + 28$

(C) $1,680t + 28$

(D) $168t$

15.

The bone mineral density of a certain person's thoracic spine was estimated to be 0.812 grams per square centimeter. What is this estimated bone mineral density in grams per square millimeter? (1 centimeter = 10 millimeters)

(A) 81.2

(B) 8.12

(C) 0.0812

(D) 0.00812

16.

A researcher surveyed undergraduate students, graduate students, and postdoctoral students. The number of undergraduate students surveyed was 7,150% of the number of postdoctoral students surveyed and the number of graduate students surveyed was 45% of the number of undergraduate students surveyed. If there were 6,435 graduate students surveyed, what was the sum of the number of undergraduate students and postdoctoral students surveyed?

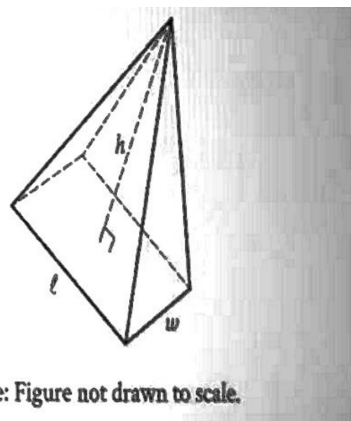


17.

The function $f(x) = \frac{1}{7}(x - 6)^2 + 4$ gives a toy car's height above the ground $f(x)$, in inches, x seconds after it started moving on an elevated track, where $0 \leq x \leq 10$. Which of the following is the best interpretation of the vertex of the graph of $y = f(x)$ in the xy -plane?

- (A) The toy car's minimum height was 4 inches above the ground.
- (B) The toy car's minimum height was 6 inches above the ground.
- (C) The toy car's height was 4 inches above the ground when it started moving.
- (D) The toy car's height was 6 inches above the ground when it started moving.

18.



The figure shown is a right rectangular pyramid, where $l = 18$ units, $w = 9$ units, and $h = 12$ units. What is the surface area, in square units, of the pyramid?

19.

For a set of three consecutive integers, the result of subtracting the greatest integer from the sum of the other two integers is 48. What is the value of the greatest of the three integers?

20.

$$h(t) = -16t^2 + b$$

The function h estimates an object's height, in feet, above the ground t seconds after the object is dropped, where b is a constant. The function estimates that the object is 40.96 feet above the ground when it is dropped at $t = 0$. How many seconds after being dropped does the function estimate the object will hit the ground?

21.

For the function f , for each increase in the value of x by c , where c is a positive constant, the value of $f(x)$ increases by a factor of 27. Which of the following equivalent forms of the function f displays $\frac{1}{c}$ as a coefficient of x ?

(A) $f(x) = 48(3)^{\frac{1}{3}x}$

(B) $f(x) = 48(3^3)^{\frac{1}{3}x}$

(C) $f(x) = 48(9)^{\frac{1}{4}x}$

(D) $f(x) = 48\left(27^{\frac{1}{3}x}\right)^{\frac{1}{2}}$

22.

In triangle XYZ , the measure of angle X is 90° . Point W lies on segment YZ , and segment WX is perpendicular to segment YZ . The length of segment WY is 524, and the length of segment WX is 393. What is the value of $\tan Z$?

A.

$$\frac{3}{5}$$

B.

$$\frac{3}{4}$$

C.

$$\frac{4}{5}$$

D.

$$\frac{4}{3}$$

23.

The solutions to $x^2 + 10x + 19 = 0$ are r and s , where $r < s$. The solutions to $x^2 + 10x + 1 = 0$ are t and u , where $t < u$. The solutions to $x^2 + 20x + c = 0$, where c is a constant, are $r + u$ and $s + t$. What is the value of c ?

24.

Hakim has a bag of 50 tiles, where 45 are labeled with a letter and 5 are blank. If Hakim selects a tile from the bag at random, what is the probability of selecting a blank tile? (Express your answer as a decimal or fraction, not as a percent.)

25.

$$\frac{1}{cx} = \frac{x}{96} + \frac{1}{c}$$

In the given equation, c is a constant. If the equation has exactly one solution, what is the value of c ?